

at frequency  $i$ ,  $E_{i,i}$  is the reference limit at frequency  $i$ ,  $c = 610/f$  ( $f$  in MHz) for occupational exposure and  $(87/f)^{1/2}$  for general-public exposure.

Assessment of exposure levels can be done either by measurement or by numerical calculations or by electromagnetic software simulations. All these methods have almost similar level of uncertainty and accuracy, depending on the method and equipment or software used.

## Radiation mitigation techniques

If the electromagnetic radiation level in any area accessible to people is higher than prescribed limits, it is strongly required to take necessary action to reduce the radiation level. To achieve the required reduction of radiation level, several mitigation techniques may be applied, either individually or simultaneously.

**1. Transmitter power reduction.** The transmitter power is directly

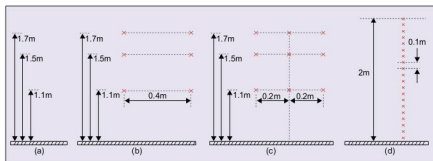


Fig. 4: Measurement points for spatial averaging

related to the power density and square of the electric field strength/magnetic field strength. So, reduction in transmitter power would result in reduction of radiation level. But, this method also leads to reduction of the coverage area.

**2. Increase in antenna height.** The power density at any observation point is the function of antenna height. If the antenna height is increased, then the power density/field strength at the observation point is reduced due to an increase in the distance to the

point of observation. The reduction of radiation level is even greater because at the same time elevation angles to the considered area are moved to another part of the vertical radiation pattern of the antenna.

**3. Decrease in VRP downtilt.** The vertical radiation pattern (VRP) of an antenna is very important for performance improvement. All energy radiated above the horizontal plane is lost, and this loss can be managed by reducing the VRP and downtilting the beam. The main beam is tilted downward in

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